

UniBank Digital Banking Platform

Comprehensive System Overview

UniBank Technology Division

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1. Executive Summary

What UniBank Is

UniBank is a full-stack digital banking platform built to deliver modern financial services across Southern Africa. It provides a mobile-first banking experience covering account management, payments, transfers, lending, bill payments, merchant services, and agent banking — all from a single integrated platform.

Key Differentiators

- **On-Premise AI:** All artificial intelligence — identity verification, document scanning, fraud detection, and a conversational banking assistant — runs entirely within the bank’s own infrastructure using an AI vision-language model. No customer data ever leaves the bank’s network, meeting POTRAZ and Reserve Bank of Zimbabwe (RBZ) data localization requirements while eliminating recurring cloud API costs.
- **Integrated EFT Switch (SynergySwitch):** A built-in electronic funds transfer switch connects POS terminals, ATMs, and national payment networks (Zimswitch) directly to the core banking system. This eliminates dependency on third-party switching providers, reduces per-transaction costs, and gives the bank full control over transaction routing and settlement.
- **Dual-Currency Architecture:** Every customer receives both a ZWG (Zimbabwe Gold) and USD account upon registration, with virtual card PANs generated automatically for each — reflecting the economic reality of Zimbabwe’s dual-currency environment.
- **Multi-Tenant White-Label Capability:** The platform can serve multiple banking institutions from a single deployment, each with isolated data, customizable branding, fee structures, and transaction limits. This opens a path to licensing the platform as a banking-as-a-service offering.
- **Comprehensive Tariff Engine:** A unified engine calculates customer fees, merchant/agent commissions, and statutory IMTT (Intermediated Money Transfer Tax at 2%) across all transaction types, ensuring regulatory compliance and transparent revenue attribution.

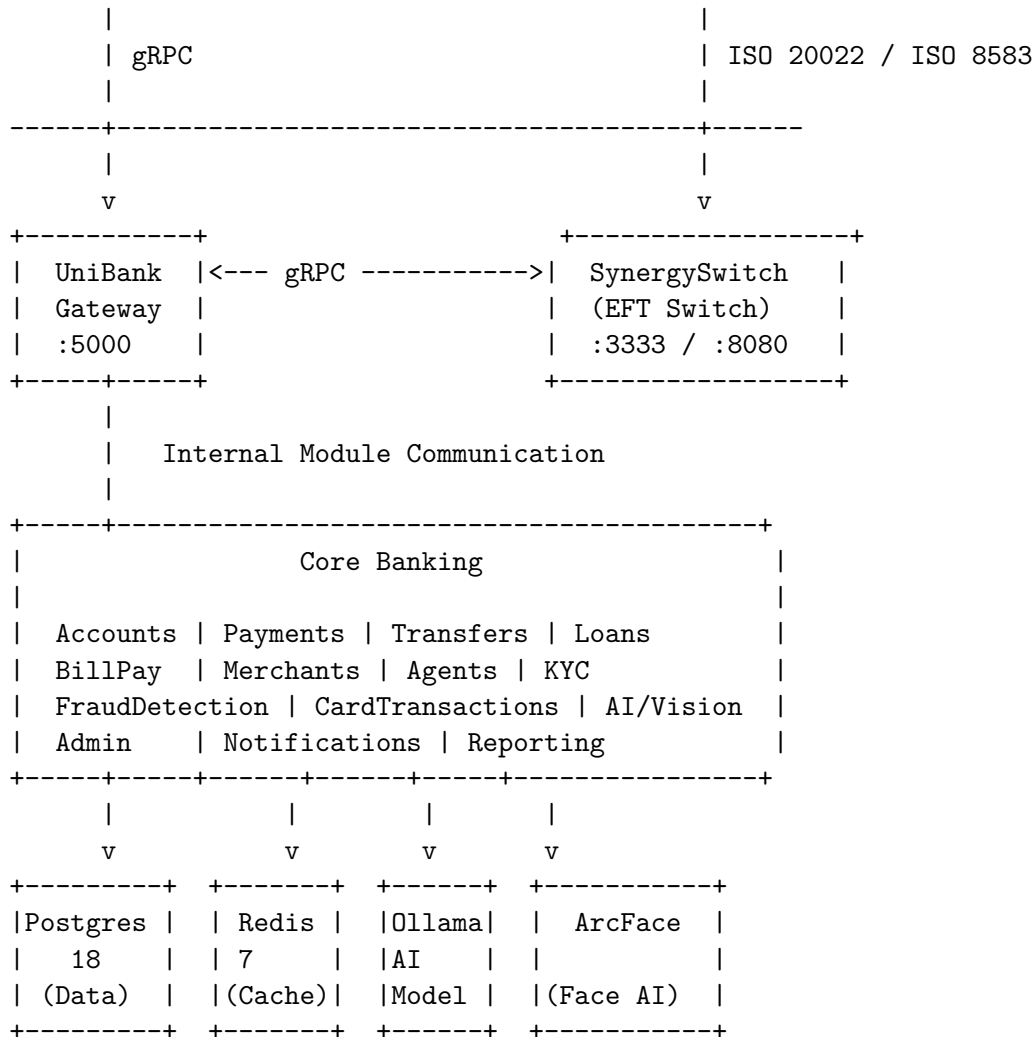
Target Market

Zimbabwe and the broader Southern African region. The platform is designed for markets where: - Credit bureau coverage is limited, requiring behavioral credit scoring - Dual-currency operations are necessary - Agent networks serve as the bridge between digital and cash economies - Regulatory requirements mandate on-premise data processing - Mobile-first adoption outpaces traditional branch banking

2. System Architecture

High-Level Component View





Monitoring: Prometheus + Grafana + Elasticsearch + Kibana

How Components Communicate

- **Mobile App to Gateway:** All communication uses gRPC, a high-performance binary protocol that is significantly faster and more bandwidth-efficient than traditional REST APIs — critical for users in areas with limited connectivity.
- **SynergySwitch Integration:** The switch accepts transactions from POS terminals and the national payment network via two industry-standard protocols (ISO 8583 over TCP for legacy systems, ISO 20022 over gRPC for modern terminals), then translates them into calls against the core banking system’s CardTransactionService.
- **Multi-Tenant Isolation:** Each tenant (banking institution) operates within its own isolated database schema. This means one institution’s customer data is completely separated from another’s at the database level, not just the application level.
- **Containerized Deployment:** All services run as isolated containers using Podman (an enterprise-grade container runtime), making deployments predictable, scalable, and independently upgradable.

Service Inventory

Service	Purpose	Resources
Gateway	Core banking API server — 14 gRPC service endpoints	512 MB
Admin Portal	Back-office web application (Blazor Server)	256 MB
SynergySwitch	EFT payment switch with BIN routing	512 MB
Terminal Manager	POS terminal lifecycle management via MQTT	256 MB
HSM	Hardware Security Module emulator — PIN encryption, MAC generation	256 MB
Notifications	Push notification delivery service	256 MB
Ollama (AI)	On-premise AI inference engine	12 GB
PostgreSQL 18	Primary relational database	512 MB
Redis 7	In-memory cache for sessions, OTPs, rate limiting	256 MB
Prometheus	Metrics collection and alerting	256 MB
Grafana	Monitoring dashboards and visualization	256 MB
Elasticsearch	Structured log storage and search	1 GB
Kibana	Log analysis dashboard	512 MB

3. Customer Journey

3.1 Registration

The registration process is designed to get customers onboarded in under two minutes:

1. **Phone Number Entry:** The customer enters their mobile number (E.164 format, validated for Southern African country codes).
2. **OTP Verification:** A 6-digit one-time password is sent via SMS, valid for 5 minutes. The OTP is stored securely in Redis and never logged.
3. **PIN Creation:** Upon OTP verification, the system automatically creates two accounts — one in ZWG and one in USD — each with a virtual card PAN (BIN prefix as defined). The customer then sets a 4-digit PIN that is hashed using BCrypt before storage. A JWT authentication token is issued.
4. **Profile Information:** The customer provides their name, date of birth, national ID number, and email address.

5. **Outcome:** The customer now has two fully provisioned accounts with KYC Level 0 (pending verification), initial daily limits of \$1,000 and monthly limits of \$5,000.

3.2 KYC Verification (Identity Verification)

KYC (Know Your Customer) verification is automated using on-premise AI, with human oversight for edge cases:

1. **ID Document Upload:** The customer photographs their Zimbabwe national ID, biometric card, or passport. The AI extracts all fields: full name, ID number, date of birth, nationality, gender, expiry date, and document type.
2. **Selfie Capture:** The customer takes a live selfie using the phone's front camera.
3. **AI Face Matching:** The Face AI neural network generates mathematical representations (embeddings) of both the selfie face and the photo on the ID document, then compares them using cosine similarity. This runs entirely on-premise using the Face AI runtime.
4. **Automated Decision:**

Face Match Score	ID Field Match	Decision
Above 60%	All fields match	Auto-Approved — KYC Level 2
40% - 60%	Partial match	Manual Review — queued for KYC Officer
Below 40%	Name mismatch	Rejected with reason provided

5. **Proof of Address (KYC Level 3):** Customers can submit a utility bill, bank statement, or lease agreement. The AI extracts the name, address, and document date, verifies the name matches, and confirms the document is recent.

3.3 Home Dashboard

After login, customers see: - **Account Balances:** ZWG and USD accounts with available balances, displayed with the last 4 digits of their virtual card PAN - **Daily Limit Usage:** Visual indicator of daily spending against configured limits - **AI Spending Insights:** 2-3 natural language insights generated by the AI analyzing recent transaction patterns (e.g., “Your grocery spending is 15% higher than last month”) - **Quick Actions:** One-tap access to Send Money, Pay Bill, Scan QR, Deposit Cheque - **Recent Transactions:** Scrollable list with counterparty names, amounts, and status - **Notification Badge:** Unread count for fraud alerts, dispute updates, and KYC status changes

3.4 Payments

NFC Tap-to-Pay: The customer taps their phone on a POS terminal. The system verifies the account, looks up the merchant, calculates the tariff (0.5% customer fee + 2% IMTT tax), checks the balance, and processes the payment in real time. For high-value transactions, a PIN prompt appears. The merchant receives the payment amount minus their discount rate commission (1.5%).

QR Code Payments: Merchants generate a QR code containing their details and the payment amount. The customer scans it with their camera, verifies the details, enters their PIN, and confirms. QR codes expire after a configurable time-to-live period to prevent replay attacks. Customer fee is 0.3% + 2% IMTT.

Bill Payments: Customers select from a list of registered bill providers (electricity, water, airtime, internet, insurance, council rates), enter their billing reference and amount, and confirm with their PIN. For prepaid utilities like electricity and airtime, a token is generated and displayed. The fee is the greater of a \$2.00 flat fee or 0.5% of the amount. Customers can save billers as favorites for quick repeat payments.

AI Bill Scanning: Instead of manually entering bill details, customers can photograph a physical bill. The AI extracts the provider name, account number, amount due, due date, and reference — automatically filling the payment form. The system even attempts to match the extracted provider to a registered bill provider in the system.

3.5 Transfers

Person-to-Person (Domestic): Send money to any UniBank customer by phone number. The system verifies both parties, validates the sender's PIN, calculates a 1% fee, debits the sender (amount + fee), and instantly credits the recipient. Both parties receive transaction records and notifications.

Cross-Border: Send money to recipients in other countries across 11 SADC corridors. The system obtains the current exchange rate, calculates a 2.5% fee, debits the sender, and initiates the outbound transfer. Estimated delivery is 1 business day for neighboring SADC countries (Zambia, Mozambique, Botswana, South Africa, and others) or 3 business days for other corridors.

3.6 Loans

The loan process is fully automated with AI-assisted verification:

1. **Eligibility Pre-Check (Optional):** The customer enters a desired amount, tenure, and purpose. The AI analyzes their account history and provides a likelihood assessment (high / medium / low), an estimated interest rate range, and a natural language assessment — before they formally apply.
2. **Application:** The customer specifies the loan amount, tenure (3, 6, 12, 18, or 24 months), and purpose, then confirms with their PIN.
3. **AI Payslip Verification (Optional):** The customer can photograph their payslip. The AI extracts employer name, employee name, gross salary, net salary, and pay period. The system compares the extracted income against any declared income, flags variances above 10%, and verifies the name matches.
4. **Credit Scoring:** The system calculates a behavioral credit score from 0 to 1,000 (detailed in Section 5).
5. **Auto-Decision:** Scores of 500 or above are auto-approved. The system calculates the interest rate, generates a full amortization schedule, and disburses the funds directly into the customer's account — all within seconds.

6. **Rejection:** Scores below 500 result in rejection with a reason provided. Customers with defaulted loans are blocked from new applications.

3.7 Disputes

When a customer identifies a problem with a transaction, they file a dispute through a guided 4-step wizard:

1. **Transaction Context:** The disputed transaction details are auto-filled and shown as read-only.
2. **Describe the Issue:** Free-text description of the problem (up to 2,000 characters).
3. **Attach Evidence:** Optional photograph of a receipt, screenshot, or other evidence (max 5 MB).
4. **AI Triage Result:** The AI automatically classifies the dispute (unauthorized transaction, duplicate charge, wrong amount, service not received, ATM failed dispensing, or card fraud), assigns a priority level (high / medium / low), routes it to the appropriate team, and provides an estimated resolution timeframe.

Customers can track their disputes through a dedicated list with filter tabs (All / Open / Investigating / Resolved) and view full details including the AI classification, assigned team, and timeline.

3.8 Fraud Alerts

The fraud detection engine continuously evaluates transactions and generates alerts (see Section 7 for details). When flagged:

1. The customer receives a push notification.
2. The fraud alert screen shows an AI-generated plain-language explanation of why the transaction was flagged, the risk level, and the triggered rules.
3. Two clear action buttons: “**This Was Me**” (confirms the transaction is legitimate) or “**Report Fraud**” (freezes the activity and automatically creates a dispute).

3.9 Agent Banking (Cash-In / Cash-Out)

Agents serve as the bridge between digital accounts and physical cash:

- **Cash-In:** An agent accepts cash from a customer and credits their digital account. The customer pays a 1% fee + 2% IMTT. The agent earns a 1.5% commission (1.0% for transactions above \$10,000).
- **Cash-Out:** A customer withdraws cash from an agent. The customer pays a 1.5% fee + 2% IMTT. The agent earns a 2.0% commission (1.5% above \$10,000).
- **Float Management:** Agents monitor their float balance and available float in real time.
- **Commission Reporting:** Agents access detailed commission reports broken down by transaction type and date range.

4. AI-Powered Features

All AI features run on-premise using two models:

Model	Purpose	Deployment
** Ollama**	Vision-language model for document understanding, chat, and analysis	Ollama container, 12 GB memory
Face AI	Dedicated face embedding model for biometric verification	Embedded runtime, in-process

Critical differentiator: No customer data — images, documents, conversations, or financial information — ever leaves the bank’s infrastructure. This satisfies Zimbabwe’s data localization requirements and eliminates the \$0.01-0.10 per-call costs of cloud AI services.

4.1 KYC Identity Verification

What it does: Compares a customer’s selfie against the photo on their ID document, and extracts all text fields from the ID.

How it works: ArcFace generates 512-dimensional face embeddings from both images, then computes cosine similarity. AI simultaneously reads the ID document and extracts structured data (name, ID number, DOB, nationality). The system performs fuzzy name matching with Levenshtein distance tolerance to handle minor OCR variations.

Business value: Reduces KYC processing time from days (manual review) to seconds (auto-approve), while maintaining quality through the three-tier decision system (auto-approve / manual review / reject).

4.2 Cheque Deposit

What it does: A customer photographs a cheque and the AI extracts all fields — cheque number, amount in figures, amount in words, payee, drawer, bank name, branch code, date, and currency.

Business value: Eliminates manual data entry for cheque deposits. The system cross-checks the amount in figures against the amount in words and flags inconsistencies.

4.3 Bill Scanning

What it does: A customer photographs a physical utility bill and the AI extracts the provider, account number, amount due, due date, reference, and currency, auto-filling the payment form.

Business value: Removes friction from bill payments — no more typing long account numbers. The system attempts to match the extracted provider name to a registered provider for one-tap payment.

4.4 Receipt Scanning

What it does: After making a purchase, a customer can photograph the receipt. The AI extracts the merchant name, date, total amount, currency, category (groceries, fuel, dining, electronics, clothing), and individual line items.

Business value: Enriches transaction records with merchant and category data, enabling more granular spending insights and better financial management tools.

4.5 Banking Assistant

What it does: A conversational AI assistant accessible via a floating action button on every screen. Customers can ask questions about their accounts, transactions, or banking features in natural language.

How it works: The assistant is injected with the customer's account context (balances, recent transactions) and uses AI to generate responses. Responses stream token-by-token for real-time conversational feel. Conversation history is maintained for the last 5 turns. Rate limited to 20 messages per hour.

Business value: Reduces call center volume by enabling self-service for common inquiries. Available 24/7 with no per-interaction cost after initial deployment.

4.6 Spending Insights

What it does: Analyzes a customer's transaction history and generates 2-3 natural language insights about their spending patterns (e.g., trends, comparisons to prior periods, unusual activity).

Business value: Increases customer engagement with the app and helps customers make better financial decisions — a feature typically reserved for premium banking apps.

4.7 Loan Eligibility Assessment

What it does: Before formally applying, customers can get an AI-powered assessment of their likelihood of approval, estimated interest rate range, and a plain-language explanation.

Business value: Reduces failed applications (which waste processing resources) and improves customer experience by setting expectations. The disclaimer clearly states this is an estimate only.

4.8 Loan Document Verification

What it does: Extracts income information from a photographed payslip (employer, employee name, gross salary, net salary, pay period) and compares it against the declared income on the loan application. Flags variances and verifies the name matches.

Business value: Automates income verification — a traditionally manual and time-consuming step in loan processing. Flags potential misrepresentation early in the process.

4.9 Dispute Triage

What it does: When a customer files a dispute, the AI reads the description and any attached evidence, then classifies the dispute type, assigns a priority level, determines the appropriate investigation team, generates a summary, and provides a confidence score.

Classification types: Unauthorized transaction, duplicate charge, wrong amount, service not received, ATM failed dispensing, card fraud.

Business value: Eliminates the initial manual triage step, ensures consistent prioritization, and routes disputes to the correct team immediately — reducing time-to-resolution.

4.10 Fraud Alert Explanation

What it does: When the fraud detection engine flags a transaction, the AI generates a plain-language explanation of why it was flagged, making the alert understandable to both customers and fraud analysts.

Business value: Customers can quickly understand and respond to alerts (confirm or report), reducing false-positive investigation workload. Analysts get clear context before beginning their review.

5. Credit Scoring Model

Why Behavioral Scoring Matters for Zimbabwe

Traditional credit scoring relies heavily on credit bureau data — payment histories across multiple lenders, credit utilization ratios, and established credit accounts. In Zimbabwe and much of Southern Africa, credit bureau coverage is limited. Many potential borrowers have no formal credit history despite being financially responsible.

UniBank’s credit scoring model addresses this gap by scoring customers based on their **observed behavior within the banking platform** — how they use their account, how actively they transact, and how they manage existing obligations. This approach:

- Includes the unbanked and underbanked population
- Rewards consistent banking behavior
- Provides real-time scoring (no stale bureau data)
- Generates revenue from a previously unserved market segment

The Scoring Algorithm

The credit score ranges from **0 to 1,000** and is composed of five weighted factors:

$$\begin{array}{ccccccccc} \text{TOTAL SCORE} & = & \text{Account Age} & + & \text{KYC Level} & + & \text{Transaction Activity} & + & \text{Loan History} & + & \text{Account Balance} \\ & & (\text{max } 200) & & (\text{max } 200) & & (\text{max } 300) & & (\text{max } 200) & & (\text{max } 100) \\ & & & & & & & & & & = 1,000 \text{ max} \end{array}$$

Factor 1: Account Age (up to 200 points)

Account Age	Points
1 year or more	200
6 months - 1 year	150
3 - 6 months	100
1 - 3 months	50
Less than 1 month	20

Rationale: Longer-standing accounts demonstrate stability and commitment to the platform.

Factor 2: KYC Level (up to 200 points)

KYC Level	Points	What It Means
Level 3 (full verification + proof of address)	200	Customer has provided comprehensive identity documentation
Level 2 (ID + selfie verified)	150	Customer identity is confirmed via AI
Level 1 (basic details)	100	Customer has provided basic personal information
Level 0 (pending)	0	Verification not yet completed

Rationale: Higher KYC levels indicate a customer willing to be fully identified — a strong indicator of legitimate intent.

Factor 3: Transaction Activity — Last 90 Days (up to 300 points) This factor considers both the number and value of completed transactions:

Transaction Count (up to 150 points):

Completed Transactions	Points
50 or more	150
20 - 49	100
10 - 19	60
5 - 9	30
Fewer than 5	10

Transaction Volume (up to 150 points):

Total Volume	Points
\$10,000 or more	150
\$5,000 - \$9,999	100
\$1,000 - \$4,999	60
\$500 - \$999	30
Below \$500	10

Rationale: Active, high-volume accounts indicate a customer who uses the platform as their primary financial tool — a reliable repayment source.

Factor 4: Loan Repayment History (up to 200 points)

Loan History	Points
Multiple loans fully repaid (paid off)	100 + 50 per loan paid (max 200)
No loan history (neutral)	100
Active loans, no defaults	80

Loan History	Points
Any defaulted loan	0

Rationale: Past loan performance is the strongest predictor of future behavior. A single default eliminates all points in this category.

Factor 5: Current Account Balance (up to 100 points)

Balance	Points
\$5,000 or more	100
\$2,000 - \$4,999	70
\$500 - \$1,999	40
\$100 - \$499	20
Below \$100	5

Rationale: A maintained balance demonstrates financial health and provides a buffer for repayment.

Score-to-Interest-Rate Mapping

Credit Score	Rating	Annual Interest Rate
800 - 1,000	Excellent	18%
650 - 799	Good	22%
500 - 649	Fair	26%
350 - 499	Poor (rejected)	30% (not offered)
0 - 349	Very Poor (rejected)	36% (not offered)

Minimum score for approval: 500

Monthly Repayment Calculation

Monthly payments are calculated using the standard amortization formula:

$$\text{Monthly Payment} = P \times [r(1+r)^n] / [(1+r)^n - 1]$$

Where: P = Principal (loan amount)

r = Monthly interest rate (annual rate / 12)

n = Tenure in months (3, 6, 12, 18, or 24)

Example: A customer with a credit score of 700 (22% annual rate) borrowing \$5,000 over 12 months would pay approximately \$466.08 per month, with a full amortization schedule generated showing the principal and interest breakdown for each payment.

Loan Processing Guardrails

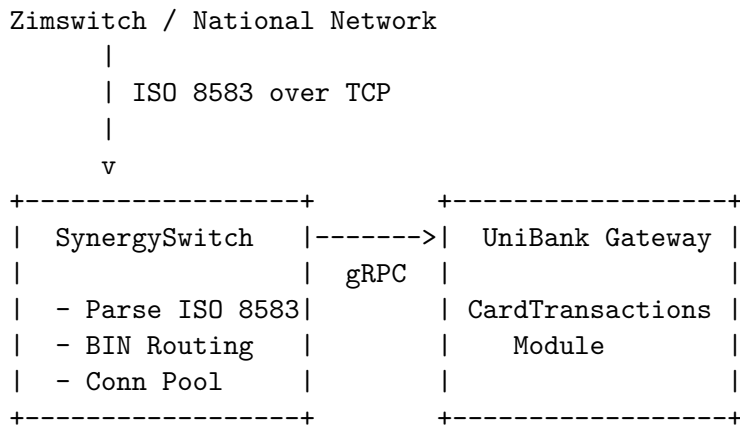
- KYC Level 1 or higher is required to apply
 - Customers with an existing defaulted loan cannot apply for a new one
 - PIN verification is mandatory
 - Approved loans are disbursed instantly to the customer's account
 - A complete amortization schedule is generated and stored at disbursement
-

6. Card Transaction Processing

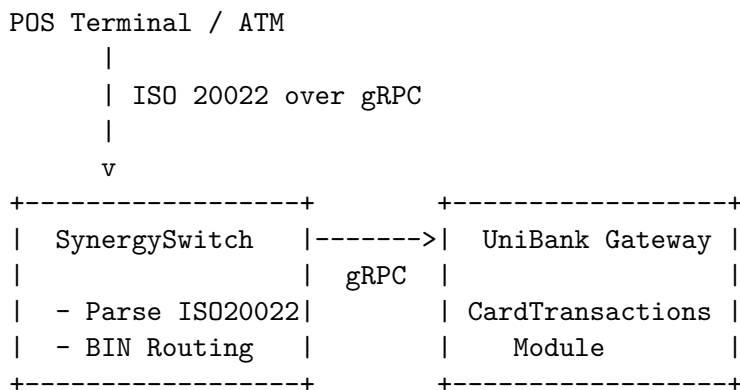
Transaction Flow

Card transactions flow through the SynergySwitch payment switch before reaching the core banking system. There are two entry paths:

PATH A - National Network (Legacy):



PATH B - Modern POS Terminals:



Both paths converge at the same CardTransactionService endpoints. Responses flow back through the same path in reverse.

How BIN Routing Works

Every payment card has a Bank Identification Number (BIN) — the first 6-8 digits of the card number. SynergySwitch maintains a routing table that maps BIN prefixes to gateways using longest-prefix matching:

1. A transaction arrives with a card number.
2. The switch checks its BIN routing table for the longest matching prefix.
3. The transaction is forwarded to the matched gateway.
4. If no specific BIN match is found, it routes to the default gateway (ordered by priority).

This routing table is cached in memory for sub-millisecond lookups and refreshed automatically when gateways or routes are modified. All changes are audit-logged.

On-Us vs. Off-Us Transactions

- **On-Us:** Both the cardholder and the merchant are UniBank customers. The system debits the cardholder's account and credits the merchant's account directly — no external settlement required. This is the most profitable transaction type.
- **Off-Us:** The cardholder is a UniBank customer but the merchant banks elsewhere. The system debits the cardholder and credits the acquiring bank's suspense account for later interbank settlement.

Supported Transaction Types

Transaction Type	Description
Purchase	Card payment at a POS terminal or online merchant
Deposit	Cash or cheque deposit at a terminal
Balance Enquiry	Card holder checks available and ledger balance
Statement Enquiry	Card holder requests a mini-statement of recent transactions

Idempotency and Reliability

Every transaction includes a STAN (System Trace Audit Number) that is checked for duplicates before processing. If a duplicate is detected (e.g., due to a network retry), the original response is returned without reprocessing the transaction — preventing double-charging.

Connection Monitoring

The switch maintains persistent gRPC channels to the UniBank gateway with: - Continuous state monitoring (Ready, Connecting, Idle, TransientFailure) - Automatic reconnection on connection drops - Keep-alive pings every 15 seconds with 5-second timeout - Prometheus metrics for all state transitions - Configurable connection pools per gateway

7. Security and Compliance

Authentication Layers

UniBank employs multiple authentication layers:

1. **Phone + OTP:** Registration requires SMS verification of phone number ownership.
2. **PIN:** A 4-digit PIN, hashed with BCrypt, is required for all sensitive operations (login, payments, transfers, loan applications, bill payments).
3. **Biometric Authentication:** Android BiometricPrompt API supports fingerprint and face recognition as a convenience layer on top of PIN authentication. If biometrics fail 3 times, the system falls back to PIN entry.
4. **JWT Tokens:** Time-limited access tokens with refresh token rotation. Access tokens expire, and refresh tokens are device-bound.

Session Security

- **Inactivity Timeout:** Configurable from 1 to 10 minutes (default: 3 minutes). After the timeout, the session locks and requires re-authentication.
- **Device Binding:** Each account is bound to a specific device ID. Transferring to a new device requires OTP verification and PIN confirmation.
- **Account Lockout:** After 5 consecutive failed PIN attempts, the account is temporarily locked. The remaining attempts counter is returned to the client.

Data Protection

- **Encrypted Storage:** All sensitive data on the mobile device is stored using Android EncryptedSharedPreferences (AES-256-GCM).
- **PIN Never Stored in Clear:** PINs are hashed with BCrypt before storage. The cleartext PIN exists only in memory during verification.
- **OTP Security:** OTPs are stored in Redis with a 5-minute expiry and are never logged.
- **Phone Number Masking:** Phone numbers are masked in all log output.

Multi-Tenant Data Isolation

Each tenant's data is isolated at the database level using PostgreSQL's schema isolation. Each tenant receives its own schema with identical table structures, ensuring: - No cross-tenant data leakage is possible at the query level - Each tenant can have independent configuration (fees, limits, branding) - Database-level security policies prevent cross-schema access - Monthly table partitioning on the transactions table ensures performance at scale

KYC Levels and Progressive Limits

KYC Level	Verification Required	Transaction Limits
Level 0	Phone + OTP only	Minimal (receive only)
Level 1	Basic profile details	\$1,000/day, \$5,000/month
Level 2	AI-verified ID + selfie	Elevated limits
Level 3	Level 2 + proof of address	Full limits

Higher KYC levels unlock higher transaction limits and access to lending products — incentivizing customers to complete verification while maintaining regulatory compliance.

Fraud Detection

The fraud detection engine evaluates every transaction against six rule types in real time:

Rule	What It Detects	Threshold
Unusual Amount	Transaction exceeding 5x the customer's 30-day average	5x average = High; 10x = Critical
Velocity Breach	Too many transactions in a short period	More than 10 per hour
Geographic Anomaly	Transaction from a different tenant/region than the account's registration	Cross-tenant mismatch
Pattern Anomaly	Same amount sent to the same recipient repeatedly	3 or more identical transfers in 24 hours
New Account Risk	Large transactions on newly created accounts	Transaction exceeding 50% of daily limit within 24 hours of creation
Failed Attempts	Multiple failed payment attempts	More than 5 failures in 30 minutes

Each alert is assigned a severity (Medium / High / Critical) and logged with full context. The fraud engine uses Redis for high-speed velocity tracking, ensuring zero impact on transaction processing latency.

Audit Trail

- All admin actions are logged with timestamps and user identification
- Gateway configuration changes are audit-logged in the switch
- All KYC review decisions record the admin user, decision, and notes
- Dispute resolution records the admin, decision, and refund amount
- Fraud alert reviews record the analyst, decision, notes, and whether the account was suspended

HSM (Hardware Security Module)

The HSM service provides cryptographic operations critical for card transaction security: - PIN block encryption/decryption (ISO Format 0, 3, and 4) - Session key derivation from master keys - MAC (Message Authentication Code) generation and verification - Payment token generation for NFC transactions

8. Admin Portal

Role-Based Access Control

The admin portal supports seven specialized roles, each with access scoped to their operational domain:

Role	Access Level	Responsibilities
Admin	Full access to all modules	System configuration, user management, complete oversight
KYC Officer	KYC module	Review identity documents, approve/reject/request resubmission
Fraud Analyst	Fraud module	Investigate fraud alerts, review flagged transactions, suspend accounts
Customer Service	Disputes + Accounts	Handle customer complaints, resolve disputes, manage account status
Loan Officer	Loans module	Review loan applications, verify AI document analysis results
Compliance Officer	Read-only access to all modules + reports	Monitor regulatory compliance, generate audit reports
Branch Manager	All modules within their branch scope	Branch-level oversight of all operations

Roles are scoped by tenant (multi-tenant isolation). Super admins have a null tenant ID, giving them cross-tenant access. Branch managers are further scoped to their specific branch.

KYC Review Workflow

1. Customer submits ID document and selfie through the mobile app.
2. AI processes the submission and makes an initial decision.
3. **Auto-approved** submissions are completed without admin involvement.
4. **Manual review** submissions appear in the KYC Officer's queue with:
 - The AI's face match score and field comparison results
 - The original selfie and ID document images
 - Extracted fields alongside the customer's profile data
5. The KYC Officer can **Approve**, **Reject**, or **Request Resubmission** with notes.

Fraud Investigation Workflow

1. The fraud detection engine generates an alert.
2. The alert appears in the Fraud Analyst's dashboard with severity, type, description, and the flagged transaction details.
3. The analyst reviews the alert and can:
 - **Dismiss** (false positive) with notes
 - **Confirm fraud** with notes, optionally suspending the customer's account
4. All review decisions are logged for audit purposes.

Dispute Resolution

1. Disputes arrive pre-triaged by AI with classification, priority, and assigned team.
2. Customer service agents see a queue filtered by status and priority.
3. For each dispute, agents see the AI's classification, confidence score, and recommended action.
4. Resolution options include refund (with configurable amount), rejection, or escalation.
5. SLA tracking ensures disputes are resolved within expected timeframes.

Reporting and Analytics

The admin portal provides comprehensive reporting:

Report	Contents
Executive Dashboard	Total users, active users, transaction count, volume, revenue, active merchants/agents/terminals, daily metrics
User Growth	Registration trends, active users, churn rate, by configurable time period
Merchant Report	Transaction volume and count per merchant, commission totals
Revenue Report	Revenue by transaction type, daily/weekly/monthly breakdowns, percentage contribution
Reconciliation	Batch-level matching between switch transactions and bank records, discrepancy identification

Reports can be exported in configurable formats with streamed delivery for large datasets.

Customer and Merchant Management

Customer Management: - Search by phone number, name, or account ID - View account details, KYC status, balances, and transaction history - Administrative actions: Suspend, Activate, Close, Freeze, Unfreeze, Reset PIN

Merchant Management: - Approve, Suspend, Activate, or Close merchant accounts - Configure commission rates and settlement frequency - View transaction history and settlement reports

Branch and User Management

- Create and manage bank branches (name, code, address, city, phone)
- Create admin users with specific roles and branch/tenant assignments
- Deactivate users with reason tracking
- List and filter admin users by role, tenant, and active status

9. Multi-Currency and Multi-Tenant Architecture

Dual-Currency Accounts

Zimbabwe's economy operates with two currencies — ZWG (Zimbabwe Gold) and USD. UniBank addresses this directly:

- **Automatic dual-account creation:** When a customer verifies their phone number, the system creates two accounts — one ZWG and one USD — in a single atomic operation.
- **Virtual card PANs:** Each currency account receives its own virtual card PAN with BIN prefix as defined, enabling card-based transactions in either currency.
- **Currency-aware transactions:** All financial operations (transfers, payments, loans) are currency-specific, with balances maintained independently per currency account.

- **Cross-currency transfers:** The exchange rate service handles conversions for cross-border transfers with real-time rate lookups.

Schema-Per-Tenant Isolation

The multi-tenant architecture uses PostgreSQL schema isolation:

```
PostgreSQL Database: unibank
|
+-- public schema (shared functions, tenant registry)
|
+-- tenant_alpha schema
|   +-- accounts
|   +-- transactions (partitioned monthly)
|   +-- loans, loan_payments
|   +-- transfers, payments
|   +-- bill_payments, bill_providers, saved_billers
|   +-- merchants, card_transactions
|   +-- fraud_alerts, fraud_rules
|   +-- kyc_documents, disputes
|   +-- admin_users
|   +-- ... (full schema)
|
+-- tenant_beta schema
|   +-- (identical structure, completely isolated data)
|
+-- tenant_gamma schema
|   +-- (identical structure, completely isolated data)
```

A provisioning function (`provision_tenant_schema`) creates the complete schema for a new tenant with all tables, constraints, indexes, and monthly partitions — enabling rapid onboarding of new banking institutions.

White-Label Capability

Each tenant can be customized through the WhiteLabel service:

Customizable Element	Example
App name	“BankX Mobile”
Logo and favicon	Custom branding assets
Color scheme	Primary, secondary, and accent colors
Support contact	Tenant-specific email and phone
Custom CSS	Additional styling overrides
Fee structure	Unique fee rules per transaction type
Transaction limits	Per-transaction, daily, and monthly limits
Commission rates	Volume-based discount tiers

This enables the bank to license the platform to other financial institutions, each presenting a fully branded experience to their customers while sharing the underlying infrastructure.

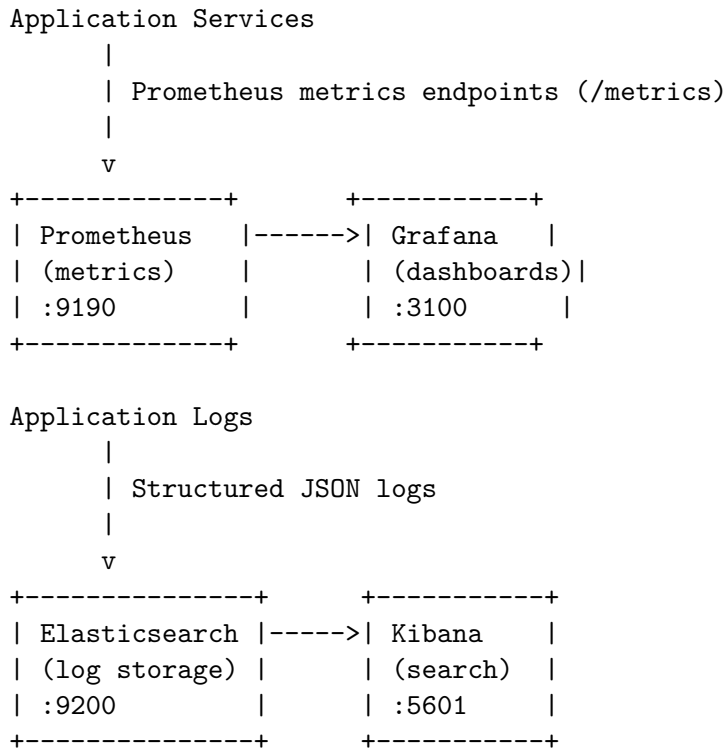
10. Infrastructure

Containerized Architecture

All services run as Podman containers on the **synergy-net** network, with deployment profiles for selective service startup:

- **“infra” profile:** PostgreSQL and Redis only (for development and testing)
- **“core” profile:** All application services (Gateway, Admin, Switch, Terminal Manager, HSM, Notifications, Ollama)
- **“monitoring” profile:** Prometheus, Grafana, Elasticsearch, Kibana

Monitoring Stack



Key metrics collected: - Transaction throughput and latency per type - Gateway connectivity state transitions - ISO 8583/ISO 20022 message counts and response code distribution - Connection pool utilization - AI inference duration and token counts - Fraud alert generation rates - Error rates across all services

Alert rules are configured in Prometheus for critical conditions such as gateway connection drops, elevated error rates, and resource exhaustion.

Service Communication

All inter-service communication uses **gRPC** (Google Remote Procedure Call), which provides: - Binary serialization (Protocol Buffers) — 5-10x smaller than JSON - HTTP/2 multiplexing — multiple requests over a single TCP connection - Bi-directional streaming — used for the chat

assistant, payment notifications, and transaction history - Strongly typed contracts — all APIs are defined in `.proto` files, preventing interface drift - Code generation — client and server code is generated automatically from proto definitions

The platform exposes **16 gRPC service definitions** covering all functionality:

Service	Endpoints	Purpose
AccountService	17 RPCs	Registration, auth, profile, balance, transactions, disputes, fraud alerts, notifications
PaymentService	6 RPCs	NFC and QR payments, tokenization, payment notifications
TransferService	2 RPCs	Domestic P2P and cross-border transfers
LoanService	4 RPCs	Loan application, details, listing, amortization schedule
BillPayService	4 RPCs	Provider listing, bill payment, saved billers
AIService	14 RPCs	KYC verification, document OCR, chat, insights, dispute triage, fraud explanation
KycService	4 RPCs	Document upload, selfie upload, verification status
AdminService	12 RPCs	Customer management, KYC review, disputes, fraud, user/branch management
MerchantService	10 RPCs	Registration, profile, transactions, settlements, commissions
AgentService	5 RPCs	Cash-in, cash-out, float management, commission reporting
CardTransactionService	4 RPCs	Purchase, deposit, balance enquiry, statement enquiry
TerminalService	3 RPCs	Terminal registration, status, firmware updates
HSMService	7 RPCs	Key generation, PIN encryption, MAC operations, tokenization
ReportingService	6 RPCs	Dashboard, growth, merchant, revenue, reconciliation, export
WhiteLabelService	4 RPCs	Branding and fee configuration per tenant

Database Architecture

PostgreSQL 18 serves as the primary data store with: - Schema-per-tenant isolation - Monthly partitioning on high-volume tables (transactions) - Automatic partition creation (3 months ahead) - Soft deletes across all entities (deleted_at column) - Check constraints for status fields and transaction types - UUID primary keys throughout

Redis 7 provides: - OTP storage with automatic expiry (5 minutes) - Session token management - Fraud detection velocity counters (1-hour and 24-hour windows) - QR payment data with time-to-live expiry - Rate limiting counters for the AI chat assistant - General-purpose caching

Scalability Considerations

The architecture is designed for horizontal scaling: - The Gateway service is stateless and can be replicated behind a load balancer - Redis provides shared session and cache state across gateway instances - PostgreSQL supports read replicas for reporting workloads - The Ollama AI service can be scaled by adding GPU-equipped nodes - The switch maintains persistent connection pools with configurable sizing - Monthly table partitioning ensures query performance doesn't degrade as data grows

This document provides a comprehensive overview of the UniBank platform as of March 2026. For technical implementation details, API specifications, or deployment procedures, please consult the engineering team.